

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Pharm. Examination

University Theory Examination May 2017

PH410 Medicinal Chemistry-IV

Date: 05.05.2017, Friday

Time: 1:30 p.m. to 04:30 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary

SECTION- I

Q 1 Attempt any TWO of the following

- A I. Enlist various drugs used as 1st line agents in the treatment of tuberculosis with their chemical structure and Mechanism of action. 05
- II. Classify antimalarial agents. Explain SAR of Cinchona Alkaloids as antimalarial agents. 05
- B I. Give classification of anticancer agents with their structure? Explain mechanism of action of antimetabolites. 05
- II. What is neoplasm? How cancer cells are differ from normal cells? Give mechanism of action of alkylating agents. 05
- C Give IUPAC name and synthesis of the following compounds: 10
- (a) Metronidazole (b) Methotrexate, (c) Sulphanilamide (d) Pyrimethamine (e) Ethambutol

Q 2 Attempt any TWO of the following

- A Why viral infections are difficult to treat? Classify antiviral agents with examples. Write a note on reverse transcriptase inhibitors. 10
- B I. What are antifungal agents? How they act? Classify with examples. 05
- II. Write a note on immunosuppressive agents. 05
- C I. Write a chemistry of anti-amoebic agents 05
- II. Explain SAR of quinolone derivatives. 05

SECTION- II

Q 3 Attempt any TWO of the following

- A What is Combinatorial Chemistry? Explain principle of it. Discuss solid support combinatorial chemistry. 10
- B I. Write a brief note on bioinformatics. 05
- II. Write a detail note on Quantitative Structure Pharmacokinetic Relationship (QSPR). 05
- C I. Explain SAR & Mechanism of action of tetracycline. 05
- II. Explain Chemistry of Aminoglycosides. 05

Q 4 Attempt any TWO of the following

- A Give chemical classification, SAR and Mechanism of action of sulfa drugs. 10
- B I. Explain SAR & Mechanism of action of chloramphenicol. 05
- II. Enlist β -lactam Antibiotics. Explain Mechanism of action & SAR of cephalosporin derivatives. 05
- C I. Chemistry of macrolide antibiotics. 05
- II. Synthesis of Ampicillin and Cephalexin. 05